

# **Pre-Deployment Normative Evaluation: An Extended Phenotype Framework for Legislative Quality Assurance**

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## **Abstract**

Legislative production is the only complex process in modern economies that operates without systematic pre-deployment evaluation. Pharmaceutical compounds undergo three-phase clinical trials before reaching patients. Engineering projects undergo stress testing before accepting loads. Financial instruments require prospectus review before public offering. Legal norms, which modify institutional substrates irreversibly and affect entire populations, enter the system without any equivalent quality filter. This paper argues that the absence of pre-deployment normative evaluation is not merely a policy gap but an evolutionary design failure: it permits memplexes with high political fitness but low institutional effectiveness to colonize the legal system and generate irreversible institutional deformation.

Drawing on Extended Phenotype Theory (EPT) and multi-level Evolutionary Game Theory (EGT), I propose a three-phase pre-deployment evaluation framework structurally analogous to pharmaceutical clinical trials: Phase I (institutional safety), measuring projected Institutional Hysteresis Rate (IHR) and identifying irreversible phenotypic effects; Phase II (normative efficacy), simulating replicator dynamics under selective pressure using the synthetic laboratory tools validated by Shapira et al. (2026); and Phase III (systemic impact), assessing interaction between the proposed norm and the pre-existing extended phenotype of the regulatory domain, measured through the Constitutional Lock-in Index (CLI) and the Institutional Evolvability Index (IEI). The framework produces five falsifiable predictions and one prospective test case (Ley 27802, Argentina, 2026) where pre-promulgation predictions were confirmed by the enacted text.

I argue that this framework is not a *de lege ferenda* proposal requiring new legislation but the institutional formalization of a duty of legislative diligence already derivable from Argentine positive law (Articles 1757-1758, Civil and Commercial Code) as applied to the process of normative production. The publication of Shapira et al. on February 23, 2026, establishing that institutional simulation tools exist and are scientifically validated, constitutes the methodological threshold that transforms the omission of pre-deployment evaluation from an excusable limitation into actionable negligence.

*Keywords: legislative quality; extended phenotype; regulatory hysteresis; institutional evolvability; pre-deployment evaluation; normative malpractice; evolutionary game theory; constitutional lock-in; synthetic laboratory; legislative diligence.*

## **I. The Quality Control Paradox in Normative Production**

### **A. Deming in the Legislature**

In 1982, W. Edwards Deming published fourteen principles for the transformation of American management. The first principle, constancy of purpose toward improvement of product and service, required that organizations define measurable outcomes before producing anything and commit to verifying whether those outcomes obtain after deployment. The third principle, cease dependence on inspection to achieve quality, demanded that organizations build quality into the process rather than relying on post-production detection of defects. By 2026, Deming's framework and its descendants (ISO 9001, Six Sigma, Total Quality Management) have penetrated every sector of productive activity in industrialized economies. Manufacturing, services, healthcare, construction, software development, financial engineering: all operate under quality assurance regimes that require specification of measurable outcomes, process controls during production, and systematic verification of results.

One productive activity remains entirely outside this framework: the production of legal norms. The Argentine Congress produces legislation without specifying measurable outcome indicators, without process controls during drafting, without pre-deployment testing, without systematic post-deployment verification, and without any institutional mechanism for tracing observed outcomes back to normative causes. The parallel with Deming's third principle is exact: the Argentine legal system relies entirely on post-production inspection (judicial review of constitutionality) rather than building quality into the legislative process itself. The judge who declares a norm unconstitutional is performing the role of the quality inspector at the end of the assembly line, the figure Deming identified as the symptom of a failed production system.

The paradox sharpens when one examines what the same legislature requires of the activities it regulates. ANMAT (Administración Nacional de Medicamentos, Alimentos y Tecnología Médica), Argentina's equivalent of the FDA, requires three-phase clinical trials before any pharmaceutical compound reaches a patient. Phase I evaluates safety in healthy volunteers. Phase II evaluates efficacy in a reduced patient population. Phase III evaluates efficacy and safety at population scale, including drug interactions, adverse effects in subpopulations, and long-term outcomes. The process can take eight to twelve years and cost hundreds of millions of dollars. The justification is straightforward: a compound that appears beneficial in laboratory conditions may produce harmful effects that only systematic evaluation can detect before the general population is exposed.

The same legislature requires structural calculations and municipal permits before any building accepts occupants. Requires CNV (Comision Nacional de Valores) prospectus approval before any financial instrument reaches investors. Requires environmental impact assessments before major infrastructure projects proceed. In each case, the logic is identical: the product interacts with a complex system (biological, structural, financial, ecological) in ways that the producer may not anticipate, and the cost of post-deployment failure exceeds the cost of pre-deployment evaluation by orders of magnitude.

Legal norms satisfy every condition that justifies pre-deployment evaluation in these other domains. They interact with a complex system (the institutional substrate). They produce effects that the producer may not anticipate (emergent equilibria, unintended institutional infrastructure, cross-domain interactions). The cost of post-deployment failure exceeds the cost of pre-deployment evaluation by orders of magnitude (as the Argentine experience with 23 failed labor reforms demonstrates). Yet they receive no pre-deployment evaluation of any kind.

## **B. The Regulator That Does Not Regulate Itself**

A manufacturer of screws in the Province of Buenos Aires must comply with IRAM standards specifying dimensional tolerances in hundredths of a millimeter. The manufacturer must document every stage of the production process. Must measure variation and maintain statistical process control charts. Must analyze root causes of defects using structured methodologies (Ishikawa diagrams, 5-why analysis, failure mode and effects analysis). Must submit to external audits by certified bodies. If the manufacturer's product causes harm because a specification was not met, the manufacturer is liable under Articles 1757-1758 of the Civil and Commercial Code for the risk created by a defective product.

The legislature that imposes these requirements on the screw manufacturer produces its own output with none of these controls. No measurable specifications of intended outcomes. No process documentation. No variation analysis. No root cause analysis. No external audit. No systematic connection between observed outcomes and the normative production process that generated them. The regulator imposes quality standards on every regulated activity except its own.

This is not merely an institutional curiosity. From the perspective of Deming's system of profound knowledge, it is a system design failure with predictable consequences. Deming's fourth principle, end the practice of awarding business based on price tag alone, translates directly to the legislative context: the political system selects norms on the basis of political fitness (coalition support, media appeal, ideological signaling) rather than institutional effectiveness (measurable outcomes, reversibility, systemic compatibility). The result is exactly what Deming predicted for any production system that optimizes for the wrong variable: high defect rates, high rework costs, declining system performance, and progressive loss of the capacity to distinguish between product variation and system failure.

## **C. The Cost Asymmetry**

The economic logic of pre-deployment evaluation rests on a cost asymmetry: the cost of testing before deployment is lower than the cost of failure after deployment. In pharmaceutical regulation, this asymmetry is well documented. The average cost of bringing a new drug to market through the full three-phase clinical trial process is estimated at 1.3 billion USD (DiMasi et al., 2016). The cost of a single post-market drug withdrawal (including litigation, settlements, regulatory penalties, and reputational damage) can exceed 10 billion USD, as the cases of Vioxx (Merck, 4.85 billion USD settlement alone) and thalidomide (incalculable human cost plus decades of litigation) demonstrate.

In normative production, the cost asymmetry is at least as severe but has never been systematically measured. Consider a single data point: Argentina's 23 failed labor reform attempts between 1990 and 2024. Each attempt consumed political capital (executive time, legislative attention, coalition-building effort, media management). Each attempt that partially succeeded before being reversed generated transitional costs (compliance adjustment, litigation, regulatory uncertainty, investment postponement). Each attempt that generated temporary institutional infrastructure left permanent residual deformation (the hysteresis measured by the IHR). The cumulative cost of these 23 failed attempts, none of which would have passed a pre-deployment evaluation using the CLI framework that predicts reform failure with 97% accuracy, represents an enormous but unmeasured loss of institutional resources.

A pre-deployment evaluation mechanism, even an imperfect one, would have flagged all 23 attempts as institutionally infeasible given the existing CLI of 0.89. The political capital consumed by those attempts could have been directed toward reforms in lower-CLI domains where success was structurally possible. The cost of the evaluation mechanism (salaries, infrastructure, methodology development) would have been a fraction of the cost of the failures it prevented. This is the same logic that justifies pharmaceutical regulation: the cost of testing is lower than the cost of deploying untested products on the population.

## **II. The Evolutionary Architecture of the Problem**

### **A. Legal Norms as Extended Phenotypes**

Extended Phenotype Theory (Dawkins, 1982) establishes that the effects of a replicator do not end at the boundary of the vehicle that carries it. A beaver's genes express themselves not only in the beaver's body but in the dam that modifies the river ecosystem for miles downstream. The dam is the extended phenotype of the beaver's genes: a material artifact built by the replicator's vehicle that modifies the selective environment in ways that favor the replicator's continued propagation.

Applied to law (Lerer, 2026b), legal norms are the extended phenotype of competing memplexes. A constitutional provision is not an abstract proposition; it is a material artifact that, once enacted, generates institutional infrastructure as tangible as a beaver's dam: specialized courts with budgets and personnel, professional communities whose economic survival depends on the norm's continued existence, academic doctrine that reproduces in law schools and scholarly journals, administrative procedures embedded in organizational routines, international treaty obligations that survive any domestic reform attempt. This infrastructure is the norm's extended phenotype, and it modifies the institutional environment in ways that favor the continued propagation of the memplex that produced it.

The EPT perspective reframes the problem of legislative quality in a precise way. The question is not whether a proposed norm is good policy in the abstract. The question is what extended phenotype the norm will generate once enacted, whether that phenotype is reversible, and how it interacts with the extended phenotypes already present in the institutional environment. These are empirical questions with measurable answers, not matters of ideological preference. And they are questions that the current legislative process does not ask.

## **B. Multi-Level Selection Conflict**

In multi-level Evolutionary Game Theory, the absence of pre-deployment normative evaluation is not an oversight but the predictable outcome of a selection conflict between three levels.

At the individual level, each legislator optimizes political fitness: reelection probability, intra-party position, media visibility, coalition standing. A legislator who proposes a norm that fails after enactment suffers less political damage than a legislator who blocks a popular proposal on the ground that it will generate irreversible institutional deformation. The political cost of action-that-fails is lower than the political cost of inaction-that-prevents-failure, because the failure materializes slowly and diffusely while the inaction is immediate and attributable. This creates a systematic bias toward enactment over evaluation at the individual level.

At the coalition level, memplexes compete for incorporation into the legal order. A memplex's fitness is measured by its replication capacity: how many minds it colonizes, how much institutional infrastructure it generates, how effectively it resists displacement by competing memplexes. A memplex that includes a pre-deployment evaluation requirement in its replication strategy is at a disadvantage relative to one that does not, because the evaluation introduces a filter that may block incorporation. Memplexes that bypass the filter replicate faster than those that accept it. Over evolutionary time, the population of enacted norms is dominated by memplexes that evaded evaluation, not by those that survived it.

At the systemic level, the institutional substrate absorbs each new norm and suffers cumulative deformations that no actor at the individual or coalition level has incentive to measure

or prevent. The IHR accumulates. The CLI rises. The IEI declines. But no actor in the legislative process bears the cost of these systemic effects directly. The cost is distributed across the entire population of regulated actors, diluted across time, and causally diffuse enough that no individual actor can trace a specific harm to a specific normative production decision. This is the classic structure of a tragedy of the commons, applied to institutional infrastructure rather than natural resources.

A pre-deployment normative evaluation mechanism would operate as a systemic-level selection filter, imposing costs on individual-level and coalition-level actors (delay, scrutiny, possible rejection) in exchange for systemic-level benefits (reduced hysteresis, preserved evolvability, avoided reform failures). The mechanism faces the same collective action problem as any commons management institution: the actors who bear the costs of the filter are not the actors who receive its benefits, and the benefits are diffuse while the costs are concentrated. Designing the mechanism to survive this selection pressure is the central challenge of institutional implementation, addressed in Section VI.

### **C. The Blind Breeder Problem**

Dawkins (1986) distinguished between artificial selection (where a breeder chooses organisms based on desired traits) and natural selection (where the environment selects without foresight). The legislative process is neither. It is what I have termed blind breeding: selection based on criteria (political fitness) that are systematically uncorrelated with the outcome variable that matters (institutional effectiveness). The breeder is not selecting for the trait that determines system performance; the breeder is selecting for a different trait entirely and assuming, without evidence, that the two are correlated.

The CLI data demonstrate that this assumption is false. Across 60 reforms in four jurisdictions, political economy variables (executive strength, union density, crisis context, ideological alignment) are statistically insignificant predictors of reform success. What predicts success is the CLI of the targeted domain: a structural property of the institutional substrate that no actor in the political process measures. Legislators are breeding for coat color and hoping that the offspring will be fast runners. The data show that coat color and speed are orthogonal.

Pre-deployment evaluation would convert blind breeding into informed artificial selection: selection based on predicted institutional outcomes rather than political fitness alone. The analogy with pharmaceutical development is precise. Drug companies do not select molecules for clinical trials based on their marketability; they select based on preclinical efficacy and safety data. Marketability is evaluated separately and later. The legislative process, by contrast, selects norms based entirely on their political marketability (coalition support, media appeal) without any preclinical evaluation of institutional efficacy and safety.

### **III. Why Hysteresis Changes Everything**

#### **A. The Teratogen Analogy**

The decisive difference between a pharmaceutical and a legal norm is not complexity, measurability, or political sensitivity. It is reversibility. A drug withdrawn from the market ceases to act on the organism. The biological system returns to its pre-exposure state (with possible residual effects that diminish over time). The withdrawal is effective because the drug's mechanism of action depends on its continued physical presence in the body.

A legal norm does not cease to act on the institutional system when it is repealed. The infrastructure it generated, specialized courts, professional communities, academic doctrine, administrative procedures, international treaty commitments, persists independently of the norm's formal validity. The deformation is permanent. The institutional system does not return to its pre-exposure state, and the residual effects do not diminish over time; they compound through path dependence, citation networks, professional socialization, and treaty lock-in.

This makes legal norms biologically closer to teratogens than to conventional pharmaceuticals. A teratogen causes permanent structural damage during a critical period of development. The damage does not reverse when the teratogen is withdrawn because the developmental window has closed: the organism has differentiated past the point where the affected structures can be rebuilt. Thalidomide's effects on limb development are the canonical example: the drug was withdrawn, but the limbs did not grow back.

The institutional analogue is precise. Argentina's Article 14bis, enacted in 1957, generated institutional infrastructure during a critical period of institutional development (the post-Peron consolidation of the welfare state). That infrastructure differentiated and compounded through four rendezvous points: the 1957 constitutional text, the 1970s judicial expansion of labor rights, the 1994 constitutional reform that elevated ILO conventions to constitutional rank, and the 2004 Vizzoti precedent that crystallized the judicial doctrine of acquired rights. Each rendezvous point added institutional layers that compounded into a CLI of 0.89, near-irreversible lock-in. The norm was never repealed, but even if it were, the infrastructure it generated (187 specialized labor courts, thousands of labor lawyers, hundreds of academic programs, dozens of incorporated international treaties, decades of binding precedent) would persist. The developmental window has closed.

#### **B. Measuring Irreversibility: The IHR in Practice**

The Institutional Hysteresis Rate (IHR) measures the permanent deformation that a norm imprints on the institutional substrate, independently of its formal validity. The IHR is not a subjective judgment about the desirability of institutional change; it is a measurement of the structural gap between the pre-norm institutional state and the post-repeal institutional state. An IHR of zero would mean that repealing the norm restores the institutional substrate to its pre-

enactment condition (no residual infrastructure). An IHR of 1.0 would mean that repealing the norm leaves the institutional substrate unchanged from its post-enactment condition (complete irreversibility).

The historical reforms database provides calibration data. Argentina's 23 labor reform attempts between 1990 and 2024 include 11 that achieved partial enactment before reversal. The time-to-reversal data exhibit a bimodal distribution: a cluster around 11 months (reforms reversed by the subsequent administration) and a cluster around 56 months (reforms reversed by judicial declaration of unconstitutionality or by the second subsequent administration). In both clusters, the IHR at reversal was strictly positive: no reversed reform left the institutional substrate in its pre-reform state. The reforms that survived 56 months before reversal left higher IHR than those reversed at 11 months, confirming that institutional infrastructure consolidates over time and that the cost of reversal increases non-linearly with duration.

This finding has a direct implication for pre-deployment evaluation: the faster a defective norm is identified, the lower the accumulated hysteresis. A Phase I evaluation that flags a projected IHR above 0.50 before enactment avoids all post-enactment accumulation. A Phase II simulation that identifies an efficacy deficit before enactment avoids the political cost of reversal and the institutional cost of transitional infrastructure. The cost-benefit calculation is not close: pre-deployment identification of normative defects dominates post-deployment correction on every measurable dimension.

### **C. The Compound Interest of Bad Law**

Hysteresis does not operate in isolation. Each norm's extended phenotype interacts with every other norm's extended phenotype in the same institutional domain. The interaction is multiplicative, not additive: a new norm enacted in a domain with high accumulated hysteresis does not simply add its own hysteresis to the existing stock; it interacts with the pre-existing infrastructure in ways that can amplify deformation exponentially.

The mechanism is Heteronomous Bayesian Updating (HBU), formalized in prior work (Lerer, 2026b). Legal professionals learn norms not by reading statutory text but by observing the reactions of other actors in the institutional environment: how judges apply provisions, how enforcement agencies prioritize, how colleagues interpret ambiguities, how academic authorities characterize the norm's meaning. Each new norm enters a learning environment already shaped by the priors that previous norms generated. The priors are not neutral; they are the accumulated product of every prior norm's extended phenotype. A new norm enacted in a domain with high accumulated hysteresis is interpreted through priors that the pre-existing infrastructure has trained into the professional population. The interpretation may diverge from the legislative text, because the priors (shaped by existing infrastructure) dominate the signal (provided by the new text) in any Bayesian updating process where the priors are strong and the signal is weak.



This is the compound interest of bad law. Each poorly designed norm not only generates its own institutional deformation; it degrades the interpretive environment in which all subsequent norms will be received. Argentina's labor regime exhibits this pattern with textbook clarity: reforms that modify peripheral provisions (registration requirements, termination procedures, payroll administration) are interpreted through priors established by the core constitutional doctrine (acquired rights, *primacia de la realidad*, *ultraactividad*), and the interpretation systematically neutralizes the reform's intended effect. The reform adds new text; the institutional environment absorbs the text into existing priors; the observed outcome is indistinguishable from the pre-reform baseline. The CLI predicts this outcome with 97% accuracy, but no actor in the legislative process performs the calculation.

## **IV. The Pharmaceutical Analogy: Structure, Translation, Limits**

### **A. What Drug Regulatory Agencies Actually Do**

The pharmaceutical clinical trial framework is not a single evaluation but a structured sequence of increasingly demanding tests, each designed to answer a specific question before proceeding to the next. Phase I (typically 20-100 healthy volunteers, 6-12 months) answers the safety question: does the compound produce unacceptable adverse effects at therapeutic doses? Phase I does not evaluate whether the drug works; it evaluates whether it is safe enough to test whether it works. Phase II (typically 100-500 patients with the target condition, 1-3 years) answers the efficacy question: does the compound produce the intended therapeutic effect in the target population? Phase II does not evaluate population-level outcomes; it evaluates whether the mechanism of action translates from preclinical models to human patients. Phase III (typically 1,000-5,000 patients, 2-4 years) answers the population question: does the compound produce net benefit at scale, accounting for drug interactions, subpopulation variation, long-term effects, and the full range of adverse events that only large samples reveal?

Three structural features of this framework matter for the legislative analogy. First, the phases are sequential and contingent: failure at any phase blocks progression to the next. A compound that fails Phase I safety never reaches Phase II efficacy testing, regardless of how promising its theoretical mechanism of action appears. Second, the criteria at each phase are pre-specified and measurable: the endpoints are defined before the trial begins, and the results are evaluated against those endpoints, not against post hoc rationalizations of observed effects. Third, the evaluation is independent of the producer: regulatory agency staff, not the pharmaceutical company's research team, assess whether the endpoints have been met. The producer has incentives to interpret data favorably; the evaluator does not share those incentives.

### **B. Translation to Normative Evaluation**

The three phases translate to normative evaluation with structural precision, though the instruments and endpoints differ. Phase I (institutional safety) answers: does the proposed norm generate irreversible institutional infrastructure? The primary instrument is the projected Institutional Hysteresis Rate (IHR). The evaluation examines six indicators, each corresponding to a dimension of institutional infrastructure that persists independently of the norm's formal validity.

The first indicator is enforcement infrastructure creation: does the norm require specialized courts, agencies, or inspectorates that demand budget, personnel, and physical plant? The second is professional community generation: does the norm create or expand professional specializations whose practitioners' economic survival depends on the norm's continued existence? The third is international treaty incorporation: does the norm activate or depend upon international obligations that survive domestic reform? The fourth is administrative procedure embedding: does the norm generate administrative routines that become standard operating procedure in public and private organizations? The fifth is doctrinal reproduction: does the norm create interpretive questions that generate academic doctrine in law schools, journals, and conferences? The sixth is precedent crystallization: does the norm create ambiguities that will be resolved by judicial precedent, adding interpretive layers that compound the original text?

Each indicator receives a score from 0 (no infrastructure generated) to 1 (maximum infrastructure generation), weighted according to the empirical patterns observed in the historical reforms database. The weighted sum produces the projected IHR. If the projected IHR exceeds 0.30, the evaluation flags the norm for enhanced scrutiny. If it exceeds 0.50, Phase II evaluation is required before the norm can proceed to legislative debate. If it exceeds 0.70, Phase III evaluation is additionally required. These thresholds are calibrated to the CLI threshold effects documented in prior work: CLI below 0.30 permits ordinary legislative reform; CLI between 0.30 and 0.50 requires crisis plus supermajority; CLI between 0.50 and 0.70 requires constitutional amendment; CLI above 0.70 requires regime-level change.

Phase II (normative efficacy) answers: does the proposed norm produce the effects declared in its statement of purposes when simulated under evolutionary dynamics? The question is precise and empirically answerable since February 23, 2026, when Shapira et al. (arXiv:2602.20021) published evidence that autonomous agent populations exhibit the same evolutionary dynamics as biological and cultural systems under selective pressure. Five operational tools described in the EPT research program constitute a functional implementation of this synthetic laboratory: JurisRank (measuring precedent fitness through citation network analysis), RootFinder (tracing doctrinal genealogies to identify ancestor memplexes), HisteresisCalc (computing projected and realized institutional deformation), IusMorfos (multi-dimensional classification of normative morphology), and LLM Council (simulating multi-agent deliberation under institutional constraints).

Phase II simulation would test the proposed norm under three conditions. First, cooperative compliance: all agents follow the declared purpose of the norm. This establishes the theoretical maximum efficacy, the outcome the proponents claim. Second, strategic exploitation: agents seek fitness-maximizing deviations from the declared purpose, including evasion through adjacent domains, formal compliance without substantive compliance, and coalition formation to capture enforcement mechanisms. This establishes the adversarial minimum, the outcome that rational actors would produce if they optimize against the norm's constraints. Third, institutional interaction: the norm operates alongside existing norms in the same domain, with agents responding to the composite incentive structure rather than to the new norm in isolation. This establishes the ecological outcome, the equilibrium that emerges when the new norm's extended phenotype interacts with existing extended phenotypes.

The gap between the cooperative compliance equilibrium (what proponents promise) and the institutional interaction equilibrium (what will actually happen) is the norm's efficacy deficit. A large efficacy deficit does not necessarily mean the norm should not be enacted; it means the proponents' declaration of intended effects is inaccurate, and the legislative body should decide based on the simulated outcome rather than the declared outcome. Pre-deployment evaluation informs the decision; it does not make it.

Phase III (systemic impact) answers: how does the proposed norm interact with the pre-existing extended phenotype of the regulated domain? The instruments are the CLI and the Institutional Evolvability Index (IEI). The CLI measures structural rigidity on a 0-1 scale: how resistant the domain is to normative change of any direction. The IEI measures adaptive capacity through four components: variation generation (V), how many alternative normative configurations the domain can produce; selection mechanisms (S), how effectively the domain filters for institutional effectiveness; constitutional constraint (C), how tightly constitutional text limits normative experimentation; and modular architecture (M), how independently different components of the domain can change without affecting others.

Together with the IHR, CLI and IEI form a diagnostic triangle. A domain with high CLI (resistant to change), high IHR (accumulating permanent deformation), and low IEI (unable to adapt) is a domain where any new norm will be absorbed into the existing infrastructure and either neutralized (if it conflicts with the dominant memplex) or co-opted (if it is ambiguous enough to be reinterpreted in the dominant memplex's favor). Argentina's labor regime occupies exactly this position: CLI 0.89, high accumulated IHR, and an IEI reflecting low modular independence between constitutional text, judicial precedent, and international treaty obligations.

### **C. Limits of the Analogy**

Two limits of the pharmaceutical analogy require acknowledgment. First, pharmaceutical clinical trials evaluate a compound's effect on biological systems that, within a trial population, are structurally similar. Legal norms operate on institutional systems that vary dramatically across

jurisdictions. A norm that succeeds in Chile (CLI 0.24) may fail in Argentina (CLI 0.89) not because the norm is different but because the substrate is different. Phase III evaluation addresses this limit directly by assessing substrate-norm interaction rather than norm quality in isolation.

Second, pharmaceutical clinical trials can use randomized controlled designs with placebo groups. Legal norms cannot be randomly assigned to populations. The synthetic laboratory partially compensates for this limitation: agent-based simulation permits controlled variation and replicated trials that observational data cannot provide. But synthetic agents are not identical to human populations, and simulation results are reduction of uncertainty rather than deterministic predictions. The appropriate epistemic stance is the same as in any modeling exercise: the simulation is useful to the extent that it reveals dynamics invisible to unaided analysis, not to the extent that it provides precise quantitative forecasts.

## **V. Institutional Design Against Capture**

### **A. The Epistemological Clergy Objection**

The most serious objection to a pre-deployment normative evaluation mechanism is democratic legitimacy. Drug regulatory agencies like the FDA derive authority from pharmaceutical expertise: the evaluation criteria are grounded in biochemistry, pharmacology, and biostatistics, disciplines with established methodologies and professional standards. Who would be the experts in a legislative FDA? The risk is creating precisely what the EPT research program identifies as an epistemological clergy (Lerer, 2025): a group that controls access to legitimated knowledge and uses that control to block institutional innovation that threatens its position.

The objection is grave, and historical experience confirms the risk. The Conseil d'Etat in France, which reviews legislation for legality before promulgation, has been criticized for embedding a technocratic worldview that systematically favors state capacity over individual liberty. The Congressional Budget Office in the United States, which evaluates fiscal impact, has been accused of using scoring methodologies that embed contestable economic assumptions as neutral technical parameters. Any pre-deployment evaluation body will face analogous pressures: the methodology embeds assumptions, the assumptions embed values, and the values may systematically favor one set of normative outcomes over another.

The resolution lies in a distinction that medical ethics resolved decades ago: the difference between process evaluation and content evaluation. A medical ethics committee that reviews clinical trial protocols does not decide whether a pharmaceutical company should develop a blood pressure medication or a cholesterol medication. It evaluates whether the development process, whatever its therapeutic objective, meets the standards of safety and efficacy that protect patients from harm. The committee is not a clergy that controls which diseases deserve treatment; it is a

quality filter that ensures all treatments, regardless of therapeutic objective, meet minimum production standards.

A normative evaluation agency would operate identically. The legislature decides what policy objectives to pursue; that is the democratic prerogative that no technical body should usurp. The evaluation mechanism verifies that the production process included the quality controls that protect the institutional substrate from irreversible damage: Was the projected IHR calculated? Was replicator dynamics simulated? Was interaction with pre-existing infrastructure evaluated? Were the results documented and publicly available? These are process questions, not content questions. A norm that scores poorly on Phase II simulation may still be enacted if the legislature determines that the policy objective justifies the institutional cost; but the decision would be informed rather than blind.

## **B. Designing Against Capture: The IEI as Institutional Blueprint**

Every regulatory agency is susceptible to capture by the interests it regulates (Stigler, 1971). A normative evaluation agency would be an immediate target of the most powerful memplexes in the legislative ecosystem. From EGT, regulatory capture is an evolutionarily stable strategy when the regulator depends on the regulated for information, funding, or career advancement. The pharmaceutical industry's documented influence on FDA advisory panels (through consulting relationships, speaking fees, and revolving-door employment) demonstrates the pattern in the domain closest to the proposed analogy.

The countermeasure is to design the evaluation agency with the properties that the Institutional Evolvability Index (IEI) identifies as protective against lock-in and capture. The IEI's four components (V, S, C, M) translate directly into institutional design specifications.

High variation generation (V) means the agency must employ evaluators with diverse disciplinary backgrounds and rotate them regularly. A body composed entirely of lawyers will embed legal assumptions; a body composed entirely of economists will embed economic assumptions. The composition should include institutional economists, evolutionary biologists, simulation specialists, legal practitioners, and domain experts relevant to the norm under evaluation. Rotation prevents the formation of stable coalitions between evaluators and the interests they evaluate.

Effective selection mechanisms (S) means the agency's methodology must be open, replicable, and subject to external verification. Every evaluation must publish its complete dataset, methodology, assumptions, and results. External researchers must be able to replicate the evaluation and challenge its conclusions. This is the scientific peer review model applied to institutional evaluation: transparency as a defense against capture.

Low constitutional constraint (C) means the agency should be created by ordinary legislation rather than constitutional amendment, so that its design can be modified as experience

accumulates. Constitutional entrenchment of the evaluation mechanism would create the same rigidity problem that the mechanism is designed to address. If the agency itself generates undesirable institutional effects, it must be reformable through the same processes it evaluates.

High modular architecture (M) means the agency's functions should be separable and independently reformable. Phase I evaluation (IHR calculation) is technically simpler and less politically sensitive than Phase II (evolutionary simulation) or Phase III (systemic impact assessment). If one module is captured or proves defective, it can be reformed without dismantling the entire mechanism. The pharmaceutical analogy holds: in the FDA's clinical trial framework, Phase I, Phase II, and Phase III evaluations are conducted by different teams with different expertise, and a failure in one phase does not compromise the integrity of the others.

### **C. Sunset Clauses as Emergency Protocol**

Laws sometimes require rapid response. Pandemic emergency measures, financial crisis interventions, national security legislation: these cannot wait for multi-year evaluation processes. Drug regulatory agencies already contemplate accelerated approval mechanisms for health emergencies: the FDA's Emergency Use Authorization, and equivalent mechanisms in other jurisdictions, permit access to unapproved compounds when no alternative treatment exists, with conditional approval allowing market access under ongoing monitoring requirements and mandatory withdrawal if post-market data reveal unacceptable risks.

The legislative equivalent is the mandatory sunset clause. A norm approved without complete pre-deployment evaluation enters into force provisionally but expires automatically at a pre-determined date unless it passes full evaluation within that period. The sunset clause limits the time window for institutional infrastructure to consolidate: if the norm is not renewed after full evaluation, the accumulated hysteresis is bounded by the sunset period rather than compounding indefinitely.

The deeper point is that the majority of laws approved with claims of urgency are not urgent. The perception of urgency is frequently an effect of the memeplex seeking incorporation into the system before the filter can operate. Speed is the parasite's strategy for evading the immune system. In biological terms, pathogens that replicate rapidly exploit the lag between infection and immune response; memeplexes that demand urgent legislative action exploit the lag between proposal and evaluation. A sunset clause neutralizes this strategy: the memeplex can achieve rapid incorporation, but it cannot achieve permanent incorporation without surviving evaluation.

Argentina's experience with emergency legislation confirms the pattern. The DNU (Decreto de Necesidad y Urgencia, executive emergency decree) mechanism has been used hundreds of times since the 1994 constitutional reform, frequently for matters that did not meet any reasonable definition of emergency. The mechanism was designed for genuine emergencies; it was captured by memeplexes that exploit the speed advantage. A sunset clause requirement for

DNU-enacted norms (automatic expiration at 180 days unless ratified by Congress after evaluation) would restore the mechanism's intended function without eliminating it.

## **VI. The Positive Law Argument**

### **A. Objective Liability for Risky Activity**

The strongest version of the argument developed in this paper is not that Argentina should create an FDA for laws. It is that Argentine law already requires, since 2026, that the legislative process meet quality controls functionally equivalent to those of pharmaceutical regulation. The only open question is whether those controls are institutionalized in a formal agency (the efficient solution) or applied diffusely through tort liability (the costly but already available alternative).

The legal basis is Articles 1757 and 1758 of the Argentine Civil and Commercial Code (2015). Article 1757 establishes that every person is liable for the damage caused by things under their custody or by risky or dangerous activities, whether by their nature, by the means employed, or by the circumstances of their performance. Article 1758 specifies that the owner and guardian of a thing, or whoever benefits from a risky activity, are jointly and severally liable for damages caused. The attribution factor is objective: it does not require proof of fault, negligence, or intent. The risk created by the activity suffices.

The argument, developed in prior work in this research program, is that the legislative process is a risky activity within the meaning of Articles 1757-1758. The risk is institutional: each norm generates extended phenotypes that may cause permanent deformation of the institutional substrate. The damage is measurable: the IHR quantifies permanent deformation, the CLI quantifies resulting rigidity, and the IEI quantifies the loss of adaptive capacity. The causal link operates through process rather than content: the negligence is not in the norm's substance but in the absence of the quality controls that a diligent production process would have included.

This structure has a critical advantage over alternative approaches to legislative liability. The Ley 26.944 (State Liability Law, 2014) requires for liability arising from lawful legislative activity a specialized individual sacrifice, a damage that is certain and actual at the time of enactment, and a causal relationship that is direct, immediate, and exclusive. These requirements exclude virtually all systemic normative damage: regulatory failures affect entire populations (no specialized sacrifice), materialize progressively (no actuality at time of enactment), and admit multiple concurrent causes (no exclusive causality). The CCyCN route through Articles 1757-1758 avoids all three barriers: the risk-based attribution does not require specialized sacrifice, does not require actuality at the time of enactment (the risk is created by the process, not by the materialization of damage), and does not require exclusive causality (joint and several liability accommodates multiple contributors to the causal chain).

## **B. The February 23, 2026 Threshold**

In medical malpractice law, the standard of care is not static. It recalibrates with each diagnostic tool that transitions from experimental to established. The physician of 1950 who did not order a CT scan was not negligent; the technology did not exist. The physician of 1985 who did not order a CT scan for every case of severe headache was not necessarily negligent; the technology was new and expensive. The physician of 2026 who does not order neuroimaging for symptoms that clearly require it is negligent: the tool exists, is accessible, and its omission falls below the standard that a diligent professional would satisfy.

The standard of care for legislative production underwent an analogous recalibration on February 23, 2026. On that date, Shapira et al. published multi-institutional evidence, methodologically reproducible and with open access to the technical infrastructure, that institutional simulation tools function under controlled conditions. The standard of diligence required of a reasonably informed legislative process changed.

Three specific omissions are now identifiable as negligence in the normative production process.

First omission: a high-impact norm enacted without a projected hysteresis impact statement. The analogue of surgery without preoperative studies. The negligence lies not in the content of the norm but in the absence of prior evaluation of the institutional infrastructure it will generate, the reversibility of that impact, and the cost of accumulated deformation on the pre-existing substrate. Applicable to any norm with a projected IHR above 0.30.

Second omission: a high-impact norm enacted without evolutionary simulation. The analogue of diagnosis without available imaging. For norms with a projected IHR above 0.50, the availability of documented simulation tools from February 2026 generates the obligation to apply them or to document the reasons for not applying them. The omission without justification is process negligence.

Third omission: a norm enacted in a domain of high accumulated hysteresis without evaluation of the pre-existing extended phenotype. The analogue of anesthesia without patient history. The Argentine labor regime, the administrative licensing system, and the social security apparatus are domains with documented accumulated hysteresis. Any reform in those domains that does not evaluate interaction with pre-existing institutional infrastructure omits information that was available and that a diligent process would have considered.

## **C. Graduated Responsibility Along the Production Chain**

Liability for normative process negligence is not uniform across the production chain. The graduation follows the degree of participation in the causal process, applying the doctrine of proportional responsibility that Argentine jurisprudence already uses for complex activities.



The president of the legislative committee who directed the treatment of a bill without commissioning an impact evaluation bears greater responsibility than the caucus legislator who followed party instructions. The executive branch that submitted the bill without reversibility analysis bears the responsibility of the product designer. The technical advisor who failed to inform the decision-makers about available tools bears independent professional responsibility. This graduation is not legal innovation; it is application of existing doctrine to a new category of risky activity.

Standing to sue should not require demonstration of a specialized individual sacrifice in the sense of *Ley 26.944*. Regulatory hysteresis affects the entire regulated population. Standing should extend to any actor who can demonstrate operation in the regulated domain and quantifiable compliance costs resulting from the institutional infrastructure generated by the norm. The measurement instruments (IHR, CLI, IEI) provide the quantification that existing standing doctrines require.

## **VII. Comparative Context**

### **A. What Existing Models Do and Do Not Do**

No existing institutional model combines all three functions proposed here. The closest precedents each address a partial dimension of the problem.

The French Conseil d'Etat reviews legislation for legality before promulgation. Its advisory opinions (*avis*) assess whether proposed norms comply with constitutional requirements, treaty obligations, and general principles of law. The review is juridical, not institutional: it asks whether the norm is legally valid, not whether it will generate irreversible institutional infrastructure or produce the declared effects under evolutionary dynamics.

The U.S. Congressional Budget Office evaluates fiscal impact. Its scoring of proposed legislation estimates budgetary costs and revenues over a ten-year window. The evaluation is financial, not institutional: it asks how much the norm will cost the federal budget, not how it will interact with the existing institutional substrate or what extended phenotype it will generate.

The OECD Regulatory Impact Assessment (RIA) framework, promoted since the 1990s and adopted by over 40 countries, evaluates costs and benefits of proposed regulation. The RIA is the closest existing model to what this paper proposes, but with a critical methodological difference: the RIA evaluates costs and benefits in static partial equilibrium (what are the expected costs and benefits at the moment of enactment, holding all else equal?), while the EPT-based framework evaluates replicator dynamics (what equilibrium will the norm converge toward as agents adapt their strategies over time?). The difference is between taking a photograph of the patient at admission and monitoring the patient's vital signs throughout treatment.

The OECD's own assessment of RIA effectiveness is mixed. Countries that implement RIA with political commitment and institutional capacity (Australia, the United Kingdom, Canada, the Netherlands) report measurable improvements in regulatory quality. Countries that implement RIA as a formal requirement without institutional capacity (several Latin American adopters, including Colombia and Mexico) report compliance without impact: the forms are filled, the boxes are checked, and the norms are enacted without modification. The difference tracks exactly the variable that the IEI measures: institutional capacity for variant evaluation. An RIA in a low-IEI environment is form without substance. The EPT-based framework makes this explicit and provides the diagnostic instruments to assess institutional readiness before implementation.

## **B. Why Argentina Specifically**

Argentina is the optimal test case for pre-deployment normative evaluation for three reasons. First, the cost of the absence of evaluation is empirically documented: 23 failed labor reforms, CLI of 0.89, IHR demonstrating permanent institutional deformation from every attempt. No other jurisdiction in the dataset provides such clean evidence of the failure mode that evaluation would address. Second, the positive law infrastructure already exists: Articles 1757-1758 CCyCN provide the doctrinal basis for process-based liability without requiring new legislation. Third, the political context is favorable: the Milei administration's deregulatory agenda creates demand for tools that distinguish between reforms that can succeed (low-CLI domains) and reforms that cannot (high-CLI domains), avoiding the waste of political capital on structurally doomed attempts.

## **VIII. The Prospective Test Case: Ley 27802**

The framework's predictive capacity is not merely retrospective. In December 2025, before the Argentine Congress voted on what would become Ley 27802 (labor modernization), five pre-promulgation predictions were published based on the CLI framework and EPT analysis of the proposed text.

Prediction 1: The reform would target provisions in the low-CLI zone (procedural requirements, registration formalities, payroll administration) while leaving intact the high-CLI constitutional infrastructure (Article 14bis acquired rights, ILO treaty obligations, Vizzoti precedent) that determines labor market outcomes. Result: confirmed. The enacted text modified registration procedures, introduced electronic payroll provisions, and created the FAL (Fondo de Asistencia Laboral) for outsourcing compliance, none of which touch the constitutional infrastructure.

Prediction 2: The reforms targeting high-CLI provisions would be drafted with sufficient ambiguity to survive legislative debate but would be neutralized by judicial interpretation applying the existing doctrinal priors. Result: partially confirmable at this stage. The text contains provisions that could be read as modifying the presumption of employment relationship (*laboralidad*), but the

language is ambiguous enough that courts applying the Vizzoti doctrine can interpret it as consistent with existing precedent.

Prediction 3: The outsourcing provisions (FAL) would generate compliance infrastructure that, once established, would resist modification regardless of the reform's broader fate. Result: confirmable by design. The FAL creates a fund administered by ARCA (formerly AFIP), requiring employer contributions, administrative procedures, and enforcement mechanisms. This infrastructure will persist even if the broader reform is reversed, consistent with the IHR prediction for norms that create administrative apparatus.

Prediction 4: The CLI for Argentina's labor domain would remain above 0.85 after the reform's enactment, because the reform does not modify the constitutional, treaty, or precedential components that dominate the CLI calculation. Result: confirmed by construction. None of the enacted provisions modify the four CLI components that account for 87% of the index value.

Prediction 5: The political narrative would frame the reform as transformative despite the structural analysis predicting marginal impact. Result: confirmed by observed media coverage and executive communication following enactment.

A pre-deployment evaluation mechanism would have produced these predictions before the reform was debated, not after it was enacted. The legislature could have proceeded with full knowledge that the reform would not modify the structural parameters that determine labor market outcomes. Whether the legislature would have chosen to proceed anyway is a political question; that it would have proceeded with accurate information rather than false expectations is the quality improvement that pre-deployment evaluation provides.

## IX. Falsifiable Predictions

If the framework is correct, the following predictions should hold. Each prediction specifies the observable outcome, the measurement instrument, the comparison condition, and the time horizon for empirical evaluation. Table 1 summarizes the six predictions; the text that follows develops each one.

**Table 1: Summary of Falsifiable Predictions**

| # | Prediction                                                                                | Instrument                                                    | Comparison                                          | Horizon                              |
|---|-------------------------------------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------|--------------------------------------|
| 1 | Evaluated norms produce lower IHR than non-evaluated norms in comparable CLI domains      | IHR difference-in-differences                                 | Evaluated vs. non-evaluated norms, same CLI bracket | 10 yr (interim 5 yr)                 |
| 2 | Phase II efficacy deficit predicts reform failure better than political economy variables | AUC comparison: efficacy deficit model vs. conventional model | Prospective sample of reform attempts               | 5 yr from first Phase II application |

|   |                                                                           |                                                                                 |                                                                 |                         |
|---|---------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------|
| 3 | Sunset clauses bound IHR accumulation below non-sunset equivalents        | Paired IHR comparison at sunset date                                            | Sunset vs. non-sunset norms, same domain and CLI                | First sunset expiration |
| 4 | High-IEI agencies resist capture longer than low-IEI agencies             | Time-to-capture (first documented evaluation reversal under political pressure) | Agencies ranked by founding IEI score                           | 15 yr                   |
| 5 | Adopting jurisdictions exhibit declining domain CLI over time             | Longitudinal CLI tracking                                                       | Adopting vs. matched non-adopting jurisdictions                 | 20 yr (interim 5 yr)    |
| 6 | HBU equilibrium shifts toward outcome-based updating in evaluated domains | Survey: information sources used by legal professionals                         | Evaluated vs. non-evaluated domains, controlling for experience | 10 yr                   |

Prediction 1 (Hysteresis reduction): Norms enacted after pre-deployment evaluation in high-CLI domains will exhibit lower realized IHR at five years post-enactment than equivalent norms enacted without evaluation in comparable domains, controlling for CLI at baseline. Measurement: IHR difference-in-differences between evaluated and non-evaluated norms in the same CLI bracket. Comparison: natural experiments arising from jurisdictions that adopt evaluation mechanisms at different times. Time horizon: ten years from first adoption, with interim measurement at five years.

Prediction 2 (Efficacy deficit predictive power): The Phase II efficacy deficit (gap between cooperative compliance equilibrium and institutional interaction equilibrium in simulation) will predict post-enactment reform failure with accuracy superior to conventional political economy variables (executive strength, union density, crisis context, ideological alignment). Measurement: AUC comparison between efficacy deficit model and conventional political economy model on a prospective sample of reform attempts. The CLI already achieves AUC of 0.97 against conventional variables; the prediction is that the efficacy deficit adds explanatory power beyond CLI alone. Time horizon: five years from first application of Phase II methodology.

Prediction 3 (Sunset clause effect): Norms approved with mandatory sunset clauses (provisional enactment pending full evaluation) will produce lower accumulated IHR at the sunset date than equivalent norms approved without sunset clauses, controlling for content similarity and domain CLI. Mechanism: the expiration date limits the time window for institutional infrastructure to consolidate, bounding the hysteresis accumulation function. Measurement: paired comparison of IHR accumulation rates between sunset and non-sunset norms in the same domain. Time horizon: varies by sunset period; initial results available at first sunset expiration.

Prediction 4 (Capture resistance): Evaluation agencies designed with high IEI properties (high V: diverse composition with rotation; high S: transparent methodology with external replication; low C: ordinary legislative creation; high M: separable phase functions) will maintain evaluation independence longer than agencies designed without attention to these properties. Measurement: time-to-capture operationalized as the first documented instance of evaluation outcome reversal following political pressure, compared across agencies with different IEI scores at founding. Time horizon: fifteen years from first adoption, consistent with the documented capture timelines for comparable regulatory agencies.

Prediction 5 (CLI trajectory): Jurisdictions that adopt pre-deployment evaluation will exhibit declining domain-specific CLI over time as the stock of poorly designed norms decreases relative to the stock of evaluated norms. Argentina's current CLI of 0.89 reflects decades of unfiltered normative accumulation. A functioning evaluation mechanism should arrest the accumulation and, over a generational time horizon, permit gradual institutional recovery. Measurement: longitudinal CLI tracking in adopting jurisdictions versus matched non-adopting jurisdictions. Time horizon: twenty years from adoption, with interim measurement at five-year intervals.

Prediction 6 (HBU interaction): In domains where pre-deployment evaluation is adopted, the HBU equilibrium will shift toward outcome-based updating (learning from observed normative effects) and away from reaction-based updating (learning from observed institutional reactions). Mechanism: when norms are accompanied by publicly available efficacy simulations, legal professionals have a new information source that competes with the traditional reaction-observation channel. Measurement: survey-based assessment of information sources used by legal professionals in evaluated versus non-evaluated domains, controlling for professional experience and institutional position. Time horizon: ten years from adoption.

## **X. Discussion**

### **A. What This Paper Does and Does Not Argue**

This paper does not argue that pre-deployment evaluation would eliminate bad legislation. No quality control system eliminates all defects; the goal is systematic reduction of defect rates below what uncontrolled production generates. Deming did not promise zero-defect manufacturing; he promised measurable, continuous improvement. The claim here is analogous: pre-deployment evaluation would produce measurable improvement in legislative outcomes, not perfection.

This paper does not argue that the proposed evaluation mechanism is politically feasible in the short term. The multi-level selection conflict described in Section II predicts that individual-level and coalition-level actors will resist any mechanism that imposes systemic-level costs on

their strategies. The argument is that the mechanism is theoretically grounded, empirically supported, legally available through existing tort doctrine, and institutionally designable. Political feasibility is a separate question that depends on contingent factors (crisis windows, coalition dynamics, leadership) outside the scope of institutional analysis.

This paper does argue three things with specificity. First, that the absence of pre-deployment normative evaluation is an evolutionary design failure with measurable consequences, not merely a policy preference. Second, that the instruments for evaluation already exist and have been validated (CLI: AUC 0.97 across 60 reforms; HBU: chi-squared 36.736,  $p < 0.0001$ ; simulation tools: independently validated by Shapira et al., 2026). Third, that Argentine positive law already provides the doctrinal basis for requiring evaluation through objective liability for risky activity, and that February 23, 2026 constitutes the methodological threshold that activates this liability.

## **B. Relationship to the Research Program**

This paper occupies a specific position in the broader EPT research program. The prior papers developed the instruments: CLI for measuring institutional rigidity (Lerer, 2025a, 2025b), IHR for measuring permanent deformation, HBU for explaining the transmission mechanism (Lerer, 2026b), AIT and GIMT for analyzing intentionality mismatch in multi-agent institutional contexts (Lerer, 2026d), PSO for identifying parasitic institutional orders (Lerer, 2026c), and the synthetic laboratory tools validated by the Shapira et al. response (Lerer, 2026a). The IEI, currently in development, completes the diagnostic triangle.

This paper integrates those instruments into a single applied framework. The three-phase evaluation structure is not an arbitrary organizational choice; it maps onto the three questions that the instruments collectively answer. Phase I (Is this norm safe?) uses IHR. Phase II (Does this norm work?) uses the synthetic laboratory. Phase III (Does this norm fit?) uses CLI and IEI. The integration reveals that the instruments, developed separately to answer distinct theoretical questions, form a coherent evaluation system when combined. That coherence was not designed into the research program; it emerged from the common theoretical foundation (EPT + EGT) that generated each instrument.

## **C. The Precautionary Principle, Inverted**

The conventional precautionary principle states that when an activity raises threats of harm, precautionary measures should be taken even if some cause-and-effect relationships are not fully established scientifically. The principle is typically invoked to justify regulation of private activity in advance of complete scientific proof of harm.

This paper inverts the application. The activity that raises threats of harm is legislative production itself. The harm is documented (23 failed reforms, CLI of 0.89, permanent institutional

deformation). The cause-and-effect relationships are established (CLI predicts failure with AUC 0.97; HBU explains the mechanism with  $p < 0.0001$ ). The precautionary measures available are pre-deployment evaluation using validated instruments. The party resisting precautionary measures is the legislature, which demands precaution from every activity it regulates but exempts itself from the same standard.

The inversion is not rhetorical. It identifies the asymmetry that makes the current institutional arrangement incoherent: the same actor that imposes the precautionary principle on regulated activities refuses to accept it for its own output. The coherence requirement, at minimum, demands that the regulator explain why its own activity should be exempt from the precautionary standard it imposes on others. No such explanation has been offered, because no one has framed the question.

## **XI. Limitations**

The quantitative instruments underpinning this framework (CLI, IHR, IEI) are theoretically motivated estimations, not fully validated empirical measurements. The CLI achieves high predictive accuracy (AUC 0.97) across the available dataset of 60 reforms in four jurisdictions, but the dataset is geographically concentrated in Latin America and Southern Europe. Extension to other constitutional traditions (common law systems, East Asian systems, post-Soviet systems) requires additional validation.

The IEI is still in development. Its four components (V, S, C, M) have been conceptually specified but not yet operationalized with the same precision as the CLI's five components. Until the IEI is fully operationalized and validated, Phase III evaluation relies primarily on the CLI, which measures rigidity but not adaptive capacity. The diagnostic triangle (CLI + IHR + IEI) is theoretically complete but empirically incomplete.

The synthetic laboratory tools have been validated for the specific purpose of simulating multi-agent institutional dynamics (Shapira et al., 2026), but their application to normative simulation is at the proof-of-concept stage. The five operational tools (JurisRank, RootFinder, HysteresisCalc, IusMorfos, LLM Council) have been applied to a limited number of case studies. Scaling to routine legislative evaluation would require infrastructure investment, methodology standardization, and systematic calibration against observed outcomes.

The positive law argument (Articles 1757-1758 CCyCN applied to legislative production) has not been tested in court. The doctrinal analysis is sound, but judicial reception is uncertain. Argentine courts have not previously applied objective liability doctrine to the legislative process, and the first cases will face political resistance regardless of doctrinal merit.

## **XII. Conclusion**

The absence of pre-deployment normative evaluation is not an oversight in institutional design. It is the predictable consequence of a multi-level selection conflict in which no actor in the legislative process has incentives to invest in systemic-level quality control. Individual legislators optimize political fitness. Memplexes optimize replication capacity. The institutional substrate absorbs cumulative deformation without feedback mechanisms. The result is a production system that would be recognized as defective in any other domain of human activity.

The argument of this paper operates at three levels. At the theoretical level, EPT and multi-level EGT explain why the absence of evaluation is stable (it is an equilibrium of the multi-level game, not an accident) and why its consequences are predictable (norms selected for political fitness rather than institutional effectiveness will generate systematic institutional deformation). At the instrumental level, the tools for evaluation exist and have been validated: CLI measures rigidity with AUC 0.97; IHR measures deformation; HBU explains the transmission mechanism; the synthetic laboratory simulates normative dynamics under controlled conditions. At the legal level, Argentine positive law already provides the doctrinal basis for requiring evaluation through objective liability for risky activity, with February 23, 2026 as the methodological threshold that transforms omission into negligence.

What remains is the institutional choice between two paths. The first path is a formal evaluation agency, an FDA for laws, designed with the IEI-derived properties that resist capture and maintain evaluation independence. This is the efficient solution: concentrated expertise, standardized methodology, systematic application. The second path is diffuse tort liability, where the evaluation obligation operates through the threat of damage claims against actors in the normative production chain who fail to apply available quality controls. This is the costly solution: case-by-case litigation, inconsistent standards, retrospective rather than prospective evaluation. But it is the solution that already exists under Articles 1757-1758 CCyCN.

Every pharmaceutical that reaches a patient has been tested for safety, efficacy, and systemic interaction. Every bridge that carries traffic has been stress-tested. Every financial instrument that reaches an investor has been reviewed for prospectus adequacy. Legal norms, which modify the institutional substrate irreversibly and affect entire populations, deserve no less. The tools are available. The legal framework exists. The question is no longer whether pre-deployment normative evaluation is feasible. It is whether the continued absence of evaluation is defensible.

*AI Disclosure: This paper was drafted with the assistance of Claude (Anthropic). All theoretical arguments, empirical claims, analytical frameworks, and errors are the author's sole responsibility. The instruments and indices described (CLI, IHR, IEI, HBU, AIT, GIMT, PSO) were developed by the author in prior work cited herein.*



## References

- Dawkins, R. (1982). *The Extended Phenotype: The Long Reach of the Gene*. Oxford University Press.
- Dawkins, R. (1986). *The Blind Watchmaker: Why the Evidence of Evolution Reveals a Universe Without Design*. W.W. Norton.
- Deming, W.E. (1982). *Out of the Crisis*. MIT Press.
- DiMasi, J.A., Grabowski, H.G. & Hansen, R.W. (2016). Innovation in the Pharmaceutical Industry: New Estimates of R&D Costs. *Journal of Health Economics*, 47, 20-33.
- Gould, S.J. & Lewontin, R.C. (1979). The Spandrels of San Marco and the Panglossian Paradigm: A Critique of the Adaptationist Programme. *Proceedings of the Royal Society B*, 205(1161), 581-598.
- Hayek, F.A. (1973). *Law, Legislation and Liberty, Vol. 1: Rules and Order*. University of Chicago Press.
- Hopman, L., Elstner, L., Avramidou, A., Prasad, S. & Lindner, D. (2026). Evaluating and Understanding Scheming Propensity in LLM Agents. *ICML 2026*. arXiv:2603.01608v1.
- Lerer, I.A. (2025a). *Constitutional Paleontology: Tracing the Genealogy of Institutional Lock-in*. Zenodo.
- Lerer, I.A. (2025b). *Constitutional Lock-in and the Phenotypic Expression of Legal Regimes: Argentina's Labor Market as Irreversible Institutional Morphology*. Zenodo.
- Lerer, I.A. (2026a). *Synthetic Chaos as Institutional Laboratory: Independent Evidence for Extended Phenotype Theory from Autonomous Agent Red-teaming (1.0)*. Zenodo. <https://doi.org/10.5281/zenodo.18777434>
- Lerer, I.A. (2026b). *Law as a Primary Adaptive Platform: Generalized Darwinism, Extended Phenotype Theory, and the Methodological Foundations of Evolutionary Jurisprudence*. Zenodo. <https://doi.org/10.5281/zenodo.18870552>
- Lerer, I.A. (2026c). *Predatory Invitations as Extended Phenotype: A Parasitic Spontaneous Order Framework for Academic Publishing (1.0)*. Zenodo. <https://doi.org/10.5281/zenodo.18853667>
- Lerer, I.A. (2026d). *Intentionality Mismatch in Commercial Liability: A Four-Level Evolutionary Analysis of Tapia Araya v. Starbucks (Argentine Supreme Court, 2026)*. Zenodo. <https://doi.org/10.5281/zenodo.18903217>
- Lerer, I.A. (2026e). *Spandrels of Accountability: Scheming Propensity Research as Inadvertent Evidence Against the Algorithmic Corporation*. Zenodo. <https://doi.org/10.5281/zenodo.18882212>
- Lerer, I.A. (2026f). *Ley 27802 as a Prospective Test of the EPT/CLI Framework: Labor Reform, Institutional Lock-in, and Predictive Validity in Argentine Law (1.0)*. Zenodo. <https://doi.org/10.5281/zenodo.18895179>
- OECD (2020). *Regulatory Impact Assessment. OECD Best Practice Principles for Regulatory Policy*. OECD Publishing.
- Shapira, N. et al. (2026). *Agents of Chaos: Exploring Failure Modes in Autonomous AI Agent Societies*. arXiv:2602.20021.

Stigler, G.J. (1971). The Theory of Economic Regulation. *Bell Journal of Economics*, 2(1), 3-21.

Tsebelis, G. (2002). *Veto Players: How Political Institutions Work*. Princeton University Press.

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## Appendix: Current Specification of the Institutional Evolvability Index (IEI)

The Institutional Evolvability Index (IEI) is the third vertex of the diagnostic triangle, alongside the CLI (measuring institutional rigidity) and the IHR (measuring permanent deformation). While the CLI and IHR are operationalized and validated, the IEI is at the specification stage. This appendix documents the current state of the instrument to enable external evaluation and replication as operationalization proceeds.

The IEI answers the question: what capacity does an institutional system have to generate viable variation, select effective adaptations, and reproduce them without losing systemic coherence? It is composed of four components, classified along the Gould axis (Gould, 2002) that distinguishes structuralist constraints (what variants are architecturally possible, regardless of their fitness) from functionalist selection (what variants survive competition within the space of possibility).

### A. The Four Components

V (Variation Generation) measures how many alternative normative configurations the domain can produce. V is structuralist in Gould's classification: it captures the architectural space of possibility, not whether the variants are functionally superior. A domain with high CLI constrains V mechanically: when constitutional text, judicial precedent, and treaty obligations restrict the space of permissible normative configurations, fewer variants can be generated regardless of political will. The tentative operationalization models V as a function of CLI:  $V = f(1 - CLI)$ . A system with CLI 0.89 has V near zero for structural reasons; a system with CLI 0.24 has V near its theoretical maximum. Validation requires measuring the number of distinct reform proposals generated per legislative period in each domain, controlling for political demand.

S (Selection Rate) measures how effectively the domain filters normative variants for institutional effectiveness rather than political fitness. S is functionalist in Gould's classification: it captures competitive survival within the space of possibility. A domain with effective selection mechanisms (evidence-based evaluation, sunset clauses, mandatory review periods) will retain variants that produce measurable improvements and discard variants that do not. A domain without selection mechanisms will retain variants based on political fitness alone, which the CLI data demonstrate is orthogonal to institutional effectiveness. Tentative operationalization: the ratio of

reforms that produce measurable outcome changes (in any direction) to total reforms enacted in the domain over a ten-year window.

C (Constitutional Constraint) measures how tightly the constitutional framework limits normative experimentation in the domain. C is functionalist: it captures the selection pressure exerted by constitutional review on normative variants. High C means the constitutional framework actively eliminates variants that deviate from established doctrine; low C means the constitutional framework permits experimentation. The distinction from CLI is that C measures the active filtering function of constitutional review, while CLI measures the cumulative structural rigidity of all institutional components. Tentative operationalization: the proportion of enacted reforms in the domain that face constitutional challenge, weighted by the success rate of those challenges, over a ten-year window.

M (Modular Architecture) measures how independently different components of the domain can change without affecting others. M is structuralist: it captures whether the domain's architecture permits localized modification or requires system-wide change for any modification. A domain with high modularity (separable registration procedures, independent enforcement mechanisms, distinct substantive and procedural components) permits incremental reform. A domain with low modularity (where modifying one provision triggers cascading effects through precedent, treaty obligations, and administrative procedures) requires all-or-nothing reform, which the CLI thresholds predict will fail in high-CLI environments. Tentative operationalization: the proportion of reform provisions that can be enacted independently (without requiring simultaneous modification of other provisions in the same domain) to total provisions in the domain.

## **B. The Gould Axis and the Diagnostic Triangle**

The IEI's four components distribute across the Gould axis: V and M are structuralist (Geoffroy-Gould dimension), capturing what variants are architecturally possible and whether the system permits localized change. S and C are functionalist (Cuvier-Darwin dimension), capturing what variants survive selection and what filtering pressure the constitutional framework exerts. Institutional evolvability depends on the proportion between both dimensions, not their sum. A system with high V and M but low S and C generates abundant variation that is never effectively filtered: reform is easy but improvement is random. A system with high S and C but low V and M filters effectively but has nothing to filter: the system is rigid but would be adaptive if it could generate variants.

The pathological case, which the IEI is specifically designed to diagnose, is the double structural blockade: V low and M low simultaneously. In this configuration, variation is noise rather than evolution. It is the inverted spandrel of Gould's framework: without architectural constraint that channels variation, the variation is functionally sterile. Argentina's labor regime exhibits this pattern: CLI 0.89, V near zero (constitutional text, precedent, and treaties restrict the space of possible variants), M near zero (modifying any provision triggers cascading effects

through the entire institutional infrastructure). The diagnostic triangle predicts that reform in this domain requires regime-level discontinuity, not incremental legislation, a prediction confirmed by the 23 consecutive failures.

### **C. Pilot Estimates**

Pilot estimates, subject to formal validation, place Chile's IEI above 0.70 (high variation generation, effective selection through regular reform cycles, low constitutional constraint on labor regulation, high modularity of the labor code) and Argentina's IEI below 0.30 (near-zero variation generation, no effective selection mechanism, high constitutional constraint through Article 14bis doctrine, near-zero modularity due to precedent-treaty-infrastructure coupling). These estimates are consistent with observed reform outcomes: Chile reformed successfully in both directions (liberalization and re-regulation) while Argentina failed at all 23 attempts regardless of direction.

Full operationalization requires: (1) formal specification of the measurement protocol for each component, (2) inter-rater reliability testing on a subset of jurisdictions, (3) validation against reform outcomes in a sample independent of the CLI validation dataset, and (4) sensitivity analysis of the component weightings. This work is planned as a dedicated paper in the EPT research program.